

1 New key establishment protocol based on random 2 walks in infinite forest.

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7 — Abstract —

8 We suggest post quantum secure protocol based on pseudorandom walk on infinite q -regular forest
9 $D(q)$ where $q = 2^m$, $m > 1$, Correspondents share positive integer n , pseudorandom tuple from $(F_q)^n$
10 and two pseudorandom input words in the alphabet F_q of length $O(1)$. They use the group of cubic
11 multivariate transformations of the vector space of points of $D(q)$ induced by walks on the forest of
12 even length as the platform for the implementation of modified Twisted Diffie-Hellman protocol of
13 Noncommutative Cryptography based on the complexity of the Conjugacy Power Problem. The
14 output of the protocol is a vector from $(F_q)^n$. In fact the action of the group on the partition set
15 of the bipartite homomorphic image $D(n, q)$ of the forest is used. Correspondents elaborate the
16 collision vector in time $O(n^2)$ in the case when the length of input words is $O(1)$ and the size of
17 private integer parameters is $O(n)$. If length of the input words is also $O(n)$ then the complexity of
18 protocol will be $O(n^3)$. We suggest also the obfuscation of the scheme with hidden graph of the
19 complexity $O(n^5)$ for which the output of the algorithm is a symbolic cubic transformation F of
20 $(F_q)^n$. It can be used symbiotically with the symmetric encryption via the adding to the ciphertext
21 from $(F_q)^n$ the vector $F(b_1, b_2, \dots, b_n)$ where the tuple $(b_1, b_2, \dots, b_n)$ of pseudorandom nature is
22 known publicly.

23 **Keywords:** Graph based cryptography, Key exchange protocols, Noncommutative
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1 Introduction.

Graph Based Cryptography is well established area of research. Most papers on this direction use graphs given by their adjacency matrices. In fact one of the first paper [1] on graph based cryptography suggested the encryption algorithm based on the pseudorandom walks on graph $D(n, q)$ defined by the system of algebraic equations and on the generalized graph $D(n, K)$ defined over the finite commutative ring K . The paper [2] suggested to use graph based dynamical systems and corresponding Chaos. Most papers on graph based Cryptography are in the area of Symmetric Cryptography (encryption algorithms, key dependent message authentication codes, generation of pseudorandom sequences of elements of fields or commutative rings [3]-[10] and [53]). Graphs $D(n, F_q)$ turned out to be very useful for the construction of efficient LDPC codes [11]-[13]. Graphs given via the system of algebraic equations turns out to be useful for the design of algorithms of Asymmetric Cryptography (see [41] and further references).

The protocol based on the Conjugacy Power Problem is well known algorithm of Noncommutative Cryptography (see [15]-[29] and recent cryptanalytic results [30]-[36]). We refer to this algorithm as twisted Diffie Hellman protocol. We suggest here its modification via the change of abstract semigroup with invertible elements on the semigroup S acting on the set M . We refer to these algorithm as Twisted Diffie Hellman protocol on transformation semigroups.

We hope that in the case when M is algebraic variety defined over commutative ring K and S acts on M as a semigroup of its polynomial or birational endomorphism will be useful for the developments of the key establishment protocols. In the paper we consider the case of affine space $M = K^n$. In this case we will use bipartite graph with the partition sets isomorphic to M and incidence relation ${}^nI(K)$ given via the system of special polynomial equations. Defined by these equations linguistic graph has natural colouring of points and lines by tuples from affine spaces K^s and K^r such that each vertex has a unique neighbour with the certain colour. In case of quadratic equations defined via bilinear form (case of Jordan-Gauss graphs) the neighbour of selected colour is defined via the system of linear equations written in row echelon form. For each linguistic graph ${}^nI(K)$ and an extension R of the commutative ring K we can consider larger linguistic graph ${}^nI(R)$ with the partition set isomorphic to R^n . So we can take $R = K[x_1, x_2, \dots, x_n]$ and consider the induced subgraph I' of ${}^nI(R)$ of all vertices with colours from $K[x_1, x_2, \dots, x_s]^s$. The walk in I' with the special starting point $(x_1, x_2, \dots, x_s, x_{s+1}, \dots, x_n)$ and destination point of kind $(f_1, f_2, \dots, f_s, f_{s+1}, \dots, f_n)$ where $f_i \in K[x_1, x_2, \dots, x_s]$, $i = 1, 2, \dots, s$, $f_j \in K[x_1, x_2, \dots, x_n]$, $j = s+1, s+2, \dots, n$ defines the polynomial map sending the tuple $(x_1, x_2, \dots, x_n)$ into $(f_1, f_2, \dots, f_n) \in K[x_1, x_2, \dots, x_n]^n$. Such maps generate the semigroup $S({}^nI(K))$ of polynomial transformations of affine space K^n . This subgroup and its conjugates with elements of $AGL_n(K)$ can serve as the platform of some algorithms of Noncommutative Cryptography. The feasibility of such algorithms heavily depends on the selected generators and the degrees and densities of their standard forms. Recall that the standard form of endomorphism F of $K[x_1, x_2, \dots, x_n]$ is the lists of monomial terms $F(x_i)$ written in the lexicographical order, the density of F is the maximal number of such terms for $F(x_i)$, $i = 1, 2, \dots, n$. The transformations from $S({}^nI(K))$ in case of linguistic graphs associated with Lie Geometries and their generalisations can be used for the constructions of public keys of Multivariate Cryptography (see [14]).

We implement Twisted Diffie Hellman protocol and its modification for transformation semigroups in the case of $S({}^nI(K))$ where $K = F_q$, $q = 2^n$, $s = r = 1$ and ${}^nI(F_q)$ is well known graph $D(n, q)$. These graphs and their connected components were introduced in [37], analogs $D(n, K)$ of $D(n, q)$ are introduced in [1]. Well defined projective limit $D(q)$ of $D(n, q)$ is a q -regular infinite forest $D(q)$ defined by the system of equations as infinite linguistic graph. The protocol implemented on the level on symbolic computations use the conjugation of $S = S(D(n, q))$ with the selected element L from $AGL_n(F_q)$. The generators are selected as special invertible cubic elements g and h of LSL^{-1} such that the subgroup $G = \langle g, h \rangle$ consists of cubic elements. Due to this remarkable property the output of this algorithm is a cubic multivariate transformation X of $(F_q)^n$. In the case when L and L^{-1} have densities of size $O(n)$ the complexity of the protocol is $O(n^5)$. In case when L

is an element of $AGL_n(q)$ the execution time will be $O(n^9)$. The security of this scheme rests on the Conjugacy Power Problem. The adversary has to solve the equation $h^x g^y h^{-x} = b$ for unknown integers x and y . This problem is NP hard in case when g and h are transformations of F_q^n given in their standard forms. The cubic transformations from the platform $LS(CD(n, q))L^{-1}$ were used in [38] for the implementation of other protocols of Noncommutative Cryptography. They have complexity $O(n^5)$ and $O(n^9)$ depending on the density of affine transformation L .

Effective implementation of the scheme modification with the use of natural action of group G on the vector space $(F_q)^n$ is possible when L is an identity transformation. This is the implementation with open graph $D(n, q)$. The collision vector has form $(d_1, d_2, \dots, d_{n-1})$. It can be used as the password for getting access to the information system.

The complexity improvement is remarkable one. This protocol requires just $O(n^2)$ elementary operation. The knowledge of the graph allows adversary to take vector p from $(F_q)^n$ and investigate the walks between p and $b(p)$ in the graph $D(nq)$ via Dijkstra algorithm. These attacks are unfeasible because the complexity of Dijkstra procedure for the graph of order v is $v \log(v)$ and the order of the graph $D(n, q)$ is $2q^n$. The analog $D(n, K)$ of $D(n, q)$ defined over arbitrary commutative ring with the unity was already used in algorithms.

So presented in the paper protocols have analogs in the case of arbitrary finite commutative ring K . Algorithms of [38] were implemented in the cases of $K = Z_q$, $q = 2^m$ and Boolean rings of order 2^m . We are going to investigate properties of suggested algorithm in cases of finite commutative rings with zero divisors in future publications.

In Section 2 we present some schemes of Noncommutative Cryptography with abstract platform semigroup or abstract semigroup of transformations. We discussed the difficulties of the implementation schemes on the base of symbolic transformations and their advantages in application to protection of encryption procedures and to access control verification algorithms. Section 3 is dedicated to technique of generating the endomorphisms of $K[x_1, x_2, \dots, x_n]$ given in terms of linguistic graphs defined by the system of algebraic equations over the commutative ring K . The protocols based on generators defined by linguistic graphs are described there. In section 4 we present the protocols for the implementation in the case of selected family of graphs of large girth $D(n, q)$ defined over the finite fields. Section 5 is dedicated to presentation of the results of computer simulations. We present the densities of the standard form of cubical transformations exchanged by correspondents. The time required for the creation of these forms is given there. In Section 6 the conclusive remarks are presented.

2 On some abstract schemes of Noncommutative Cryptography.

Everybody knows that Diffie Hellman key exchange protocol can be formally considered in general case of any finite group or semigroup G . In the case of group corresponding El Gamal form cryptosystem can be introduced. Notice that security of this algorithm depends not only on abstract group G but on the way of its generation in computer memory. For instance if $G = Z_p^*$ is multiplicative group of large prime field then discrete logarithm problem (DLP) is the base for classical Diffie-Hellman key exchange protocol, if the same abstract group is given as additive group of Z_{p-1} protocol is insecure because DLP will be given by linear equation. The results on the discovery of polynomial time solutions of factorization and discrete logarithm problems (d. l. p.) with the use of theoretical quantum computer obtained by Peter Shor lead to intensive research on Postquantum Cryptography and Noncommutative Cryptography which studies protocols in terms of noncommutative semigroup and its several elements in particular.

If G is noncommutative group correspondents can use conjugations of elements involved in protocol, some algorithms of this kind were suggested in [4], [5], [6], [7], where group G is given with the usage of generators and relations. Security of such algorithms is connected with Conjugacy Search Problem (CSP) and Power Conjugacy Search Problem (PCSP), which combine CSP and Discrete Logarithm Problem and their generalisations. This direction belongs to Non-commutative cryptography which is active area of cryptology, where the cryptographic primitives and systems are

based on algebraic structures like groups, semigroups and noncommutative rings. This direction of security research has very rapid development. One of the earliest applications of a non-commutative algebraic structure for cryptographic purposes was the use of braid groups to develop cryptographic protocols. Later several other noncommutative structures have been used. Latest cryptanalytic results for groups defined via generators and relations method can be found in [30]-[36]. These methods are not applicable to the situation when elements of the platform group or semigroup are given via their standard form of polynomial transformations of affine spaces of kind K^n . Some implemented protocols used subgroups or subsemigroups of such transformations. These algorithms do not use the natural action of affine Cremona semigroup of endomorphisms of $K[x_1, x_2, \dots, x_n]$ on the free module K^m , they use just the semigroup operation (see [38]-[41]).

Let us consider the scheme of Twisted Diffie-Hellman protocol for the semigroup S of transformation acting on the set M . Alice and Bob share the mutually inverse element h and h^{-1} and element g from S . Additionally they share some element m from M . Alice selects his positive integer $k(A)$ and integer $r(A)$ and computes the element $h^{r(A)}g^{k(A)}h^{-r(A)} = g(A)$. She sends semigroup element $g(A)$ to Bob. Bob chooses his positive parameter $k(B)$ and integer $r(B)$. He forms $h^{r(B)}g^{k(B)}h^{-r(B)} = g(B)$. and sends it to Bob. Assume for the simplification that $r(A)$ and $r(B)$ are positive. Alice takes element m and computes $m(A)$ from M via consecutive application of h exactly $r(A)$ times. So she computes 1m_A as $h(m)$, 2m_A as $h({}^1m_A)$, $\dots$, $m(A) = {}^{r(A)}m_A$ as $h({}^{r(A)-1}m_A)$. Next she takes transformation $g(B)$ and applies it to $m(A)$ exactly $k(A)$ times. So she gets 1b_A as $g(B)(m(A))$, 2b_A as

$$g(B)({}^1b_A), \dots, {}^{k(A)}b_A = g(B)({}^{k(A)-1}b_A) = b(A).$$

Finally Alice computes the collision element $c = c(A)$ as

$$h^{-1}(b(A)) = {}^1c(A), h^{-1}(c(A)) = {}^2c(A), \dots, h^{-1}({}^{r(A)-1}c(A)) = c(A).$$

Bob takes element m and computes $m(B)$ from M via consecutive application of h exactly $r(B)$ times. So he computes 1m_B as $h(m)$, 2m_B as $h({}^1m_a)$, $\dots$, $m(B) = {}^{r(B)}m_B$ as $h({}^{r(B)-1}m_B)$. Next he takes transformation $g(A)$ and applies it to $m(B)$ exactly $k(B)$ times. So he gets 1b_B as $g(A)(m(B))$, 2b_B as

$$g(A)({}^1b_B), \dots, {}^{k(B)}b_B = g(A)({}^{k(B)-1}b_B).$$

Finally Bob computes the collision element $c(B)$ as

$$h^{-1}(b(B)) = {}^1c(B), h^{-1}(c(B)) = {}^2c(A), \dots, h^{-1}({}^{r(B)-1}c(B)) = c(B) = c.$$

2.1 Options for adversary

Adversary can work on the level of abstract semigroup and try to solve the equation $H^xG^yH^{-x} = R$ where $R = g(A)$ or $g(B)$ for variables x and y . Additionally he/she can use pairs of kind $(m, R(m))$ to investigate equations $H^x(m)G^y(m)H^{-x}(m) = R(m)$.

Presented above scheme can be considered in the case when elements g and h are taken from affine Cremona semigroup ${}^nCS(K) = \text{End } K[x_1, x_2, \dots, x_n]$. Endomorphisms g and h of $K[x_1, x_2, \dots, x_n]$ are given via their values $g(x_i)$ and $h(x_i)$ on variables. These values are given by their standard forms of multivariate polynomials, i.e. lists of monomial terms ordered lexicographically. Recall that each element F of ${}^nCS(K)$ given by $(f_1(x_1), f_2(x_2), \dots, f_n(x_n))$ induces the transformation ${}^tF : x_1 \rightarrow f_1(x_1, x_2, \dots, x_n), x_2 \rightarrow f_2(x_1, x_2, \dots, x_n), \dots, x_n \rightarrow f_n(x_1, x_2, \dots, x_n)$. The homomorphism sending F to tF may have some nontrivial kernel. In this case we refer to the protocol as twisted Diffie-Hellman protocol on affine space. The collision vector of such protocol is a tuple of kind $(a_1, a_2, \dots, a_n)$ where $a_i \in K$.

Note that abstract twisted Diffie-Hellman protocol can be used for the subsemigroup $S = \langle g, h \rangle$ of affine Cremona semigroup ${}^nCS(K)$ as well. In this case after the generation of $g(A)$ and $g(B)$ Alice computes the collision endomorphism $C = H^{r(A)}g(B)^{k(A)}H^{-r(A)}$ of $K[x_1, x_2, \dots, x_n]$ and Bob gets C as element $C = H^{r(B)}g(A)^{k(B)}H^{-r(B)}$. So C is given via the tuple of multivariate

polynomials $C(x_i), i = 1, 2, \dots, n$ written in its standard form. We refer to these algorithms as twisted Diffie-Hellman protocols on symbolic mode. Main problem for the implementation of both schemes in the case of elements from affine Cremona semigroup ${}^nCS(K)$ is a high cost of the computation of the product of two elements.

The degree $\deg F$ of endomorphism F is defined as the maximum of degrees of multivariate polynomials $F(x_i), i = 1, 2, \dots, n$. We define the density of multivariate polynomial f as its number of monomial terms. We define the density of F as the maximal value of $F(x_i)$. The density of an endomorphism F in 'general position' of degree d is of size $O(n^d)$. Assume that F and G are nonlinear endomorphisms of degrees $d, d \geq 2$ and $e, e \geq 2$ are in general position. Then the substitution of the expressions $F(x_i)$ in the monomial term of G requires $O(n^{de})$ elementary operations. The number of all monomial terms of G is $O(n^{e+1})$. Thus the complexity of computation the composition of F and G is $O(n^{de+e+1})$. The composition of $O(n)$ nonlinear endomorphisms is unfeasible because the degree of their product is evaluated as the product of degrees of endomorphisms, resulting element will be of exponential degree in variable n .

All mentioned implementations of the twisted Diffie-Helman protocol on symbolic map use the discovery of the stable subsemigroup, i.e. subsemigroups such that the maximal degree of its representatives is d where d is some constant. For instance in [41] the stable subsemigroups of degree 2 were used, the paper [38] based on the cubic stable semigroups.

The importance of the protocols on platforms of multivariate transformations can be justified by the following remarks.

► **Remark 1.** The input-output scheme of the protocol of Noncommutative Cryptography based on the symbolic computations with the endomorphisms of $K[x_1, x_2, \dots, x_n]$ is the following one.

Correspondents Alice and Bob have publicly known elements $g_1, g_2, \dots, g_k$ from the affine semigroup ${}^nCS(K)$ and elaborate the nonlinear collision map F in n variables from ${}^nCS(K)$. Assume that they use symmetric encryption algorithm with the plainspace K^n , Before sending the ciphertext $c = (c_1, c_2, \dots, c_n)$ they take publicly known $b = (b_1, b_2, \dots, b_n) \in K^n$ of pseudorandom or genuinely random nature and the correspondent sends $F(b) + c$ instead c .

Cryptanalytic tools for breaking such schemes without the break of the protocol are unknown. Notice that in the case when ciphertext coincides with the plaintext we obtain one time pad encryption with the password hidden via endomorphism of affine Cremona semigroup. Other idea is to use some simple publicly known deformation rule from the affine Cremona semigroup in the form $g(y) \in K[y]$ and use $g(F(x_1, x_2, \dots, x_n))$.

► **Remark 2.** Assume that Bob is public user who ask the system administrator Alice to enter the information system. So they execute the protocol, Bob sends pseudorandom b to Alice. She sets the $F(b_1, b_2, \dots, b_n)$ (or $g(F(b_1, b_2, \dots, b_n))$) to enter the system.

► **Remark 3.** Notice that twisted Diffie-Hellman protocol on affine space K^n can be also used for the protection of symmetric encryption with the plainspace of K^n . In fact generation of $G(A)$ and $G(B)$ is independent from the choice of $d = (d_1, d_2, \dots, d_n)$. Before the start of communication session Alice and Bob execute the computation and exchange of these transformations.

Then each time before sending the ciphertext correspondents set d as pseudorandom b and compute the collision vector b' of the protocol with their private integer parameters. Correspondent sends $b' + c$ to his/her partner. Sender can use $g(b')$ where g is the deformation rule.

In the case of twisted Diffie-Hellman protocols on affine spaces we can use alternative approaches for the presentation of the endomorphism of $K[x_1, x_2, \dots, x_n]$ or corresponding to it multivariate transformation of the affine space.

One of such approaches based on linguistic $I(K)$ was briefly presented in the previous section. The formal definition of linguistic graph and its application for the implementation of the protocols is presented in the next section.

3 On symbolic computations with linguistic graphs

Linguistic graph of type s, r, m defined over the commutative ring K is a bipartite graph which partition sets are affine spaces of kind K^{s+m}, K^{r+m} and the incidence relation is given via the system of nonlinear algebraic equations over the commutative ring K with unity (see [38] and further references). We identify points with tuples of kind $(x) = (x_1, x_2, \dots, x_s, x_{s+1}, x_{s+2})$ and lines with tuples $[y] = [y_1, y_2, \dots, y_r]$. Brackets and parenthesis are convenient to distinguished type of the vertex of the graph. Elements (x) and $[y]$ are incident $(x)I[y]$ if and only if the following relations hold.

$$\begin{aligned} a_1 x_{s+1} - b_1 y_{r+1} &= f_1(x_1, x_2, \dots, x_s, y_1, y_2, \dots, y_r), \\ a_2 x_{s+2} - b_2 y_{r+2} &= f_2(x_1, x_2, \dots, x_s, x_{s+1}, y_1, y_2, \dots, y_r, y_{r+1}), \\ &\dots, \\ a_m x_{s+m} - b_m y_{r+m} &= f_m(x_1, x_2, \dots, x_s, x_{s+1}, \dots, x_{s+m-1}, y_1, y_2, \dots, y_r, y_{r+1}, \dots, y_{r+m-1}) \end{aligned} \quad (1)$$

where a_j and b_j , $j = 1, 2, \dots, m$ are not zero divisors, and f_j are multivariate polynomials with coefficients from K . We refer to the tuple $(x_1, x_2, \dots, x_s)$ of the point

$$x = (x_1, x_2, \dots, x_s, x_{s+1}, x_{s+2}, \dots, x_{s+m})$$

as the colour $\rho(x)$ of the point x . We say that The tuple $(y_1, y_2, \dots, y_r)$ is the colour $\rho([y])$ of the line $[y_1, y_2, \dots, y_r, y_{r+1}, \dots, y_{r+m}]$.

We refer to K^s and K^r as the colour spaces of the linguistic graph. For each point p and the colour a from the colour space K^r there is a unique line $N_a(p)$ which is incident to p . For each line $[l]$ and the colour a from the colour space K^s there is a unique point $N_a(l)$ which is incident to y . We introduce the operator $N_a(x)$ on the vertices of the graph which value coincides with $N_a(p)$ if x is a point p and with $N_a(l)$ if $[l]$ is the line.

The walk in the graph $I(K)$ is the sequence of vertices $v_1, v_2, \dots, v_k$ such that $v_i I v_{i+1}$. The information on the walk can be given via the starting point v_1 and sequence of colours $a_2, a_3, \dots, a_k$ of vertices v_i , $i = 2, 3, \dots, k$.

Let K' be the extension of commutative ring K . We define new graph $K' I(K)$ as linguistic graph with partition sets $(K')^{s+m}$ and $(K')^r + m$ such that the incidence between point $(x_1, x_2, \dots, x_{s+m})$ and line $[y_1, y_2, \dots, y_{r+m}]$ is given by equations (1) with the same coefficients a_i and b_i from K and same polynomials f_i from $K[x_1, x_2, \dots, x_{s+m}, y_1, y_2, \dots, y_{r+m}]$. In particular we can take infinite ring $R = K[x_1, x_2, \dots, x_{s+m}]$ as the extension of K to define graph $R I(K)$. We can take the special point $(x_1, x_2, \dots, x_{s+m})$ as the starting point of the walk w of even length defined by colours ${}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB$ where iA are elements of $K[x_1, x_2, \dots, x_s]^s$ and ${}^iB \in K[x_1, x_2, \dots, x_s]^r$. So ${}^iA = ({}^iA_1, {}^iA_2, \dots, {}^iA_s)$, ${}^iB = ({}^iB_1, {}^iB_2, \dots, {}^iB_r)$ where iA_j and iB_j are elements of $K[x_1, x_2, \dots, x_s]$.

Let u_w be the destination point of the walk w . Then w is the tuple

$$({}^k B_1, {}^k B_2, \dots, {}^k B_s, g_{s+1}, g_{s+2}, \dots, g_{s+m})$$

where $g_i \in K[x_1, x_2, \dots, x_{s+m}]$. We can identify this tuple ${}^{m+s}\Psi({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB) = {}^n\Psi(w)$ with the corresponding endomorphism of R , i.e. element of affine semigroup ${}^m + {}^sCS(K)$. Note that elements of kind iAs can be identified with elements of ${}^CS(K)$. So we can calculate their products. We define an action of iA on jB as

$$\begin{aligned} ({}^jB_1({}^iA(x_1), {}^iA(x_2), \dots, ({}^iA(x_s))), {}^jB_2({}^iA(x_1), {}^iA(x_2), \dots, ({}^iA(x_s))), \\ \dots, {}^jB_r({}^iA(x_1), {}^iA(x_2), \dots, ({}^iA(x_s)))) \end{aligned}$$

and denote it as ${}^iA({}^jB)$ (see [54]).

Let us consider the totality ${}^{s,r}F(K)$ of tuples of kind $w = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)$ of various length k where ${}^iA \in K[x_1, x_2, \dots, x_s]^s$, ${}^iB \in K[x_1, x_2, \dots, x_s]^r$. We refer to these tuples as symbolic

walks. Assume that ${}^1w = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)$ and ${}^2w = ({}^1C, {}^1D, {}^2C, {}^2D, \dots, {}^lC, {}^lD)$ are two elements of ${}^{s,r}F(K)$. We define the product ${}^1w \times {}^2w$ as the tuple

$$({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB, {}^kB({}^1C), {}^kB({}^1D), {}^kB({}^2C), {}^kB({}^2D), \dots, {}^kB({}^lC), {}^kB({}^lD)).$$

This product allows us to consider ${}^{s,r}F(K)$ as the semigroup.

► **Lemma 4.** [38]. *The map Ψ from ${}^{s,r}F(K)$ to ${}^nCS(K)$ sending symbolic word w to ${}^m + s\Psi(w)$ is the homomorphism of semigroups.*

Note that the information on $G = {}^m + s\Psi(w)$ can be given as its standard form or alternatively as the symbolic word w and the graph $I(K)$.

Let us consider the symbolic word $w = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)$ with the kB from affine Cremona group ${}^C S(K)$. So the inverse ${}^kB^{-1}({}^{k-1}B), {}^kB^{-1}$ is well defined. We form the reverse word w^* of w as the tuple

$$({}^kB^{-1}({}^kA), {}^kB^{-1}({}^{k-1}B), {}^kB^{-1}({}^{k-1}A), \dots, {}^kB^{-1}({}^1B), {}^kB^{-1}({}^1A), {}^kB^{-1}).$$

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► **Lemma 5.** [38]. *Let w be symbolic word $w = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)$ and ${}^kB \in {}^sCG(K)$. Then ${}^s + m\Psi(w)$ is invertible and ${}^s + m\Psi(w^*)$ is its inverse.*

3.1 Protocol 1

Alice and Bob share the information on the linguistic graph $I(K)$ and two symbolic walks of kind ${}^1w = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)$ and ${}^2w = ({}^1C, {}^1D, {}^2C, {}^2D, \dots, {}^lC, {}^lD)$. Assume that ${}^kB \in {}^sCS(K)$. To define lD we consider subset $J = i(1)i(2), \dots, i(j), j \leq s$ and the tuple $(x_1, x_2, \dots, x_s)$ and specialise its variables $x_i(t)$ as constant $b(i(t))$ or selected variable x_i , where j is not an element of J . In case of such choice the endomorphism lDI is idempotent, i.e. $({}^lD)^2$ coincides with lD . Additionally they share the point $p = (p_1, p_2, \dots, p_s, p_{s+1}, \dots, p_{s+m})$ of $I(K)$ with pseudorandom or genuinely random coordinates.

Alice selects her private keys as positive integers $r(A)$ and $k(A)$. She computes

$$w(A) = ({}^1w)^{r(A)}({}^2w)^{k(A)}({}^1w^*)^{r(A)}$$

in the semigroup $F(K)$. Secondly Alice computes the transformation ${}^m + s\Psi(w(A)) = G(A)$ in its standard form and sends it to Bob.

He selects his private keys as positive integers $r(B)$ and $k(B)$. Bob computes $w(B) = ({}^1w)^{r(B)}({}^2w)^{k(B)}({}^1w^*)^{r(B)}$ in the semigroup $F(K)$. He computes the transformation ${}^m + s\Psi(w(B)) = G(B)$ in its standard form and sends it to Alice.

Noteworthy that $G(A)(x_i)$ and $G(B)(x_i)$ for $i = 1, 2, \dots, s$ coincides with ${}^lD(x_i)$. Alice takes the word

$${}^1w^{r(A)} = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)^{r(A)} = ({}^1\alpha, {}^1\beta, {}^2\alpha, {}^2\beta, \dots, {}^{r(A)}k\alpha, {}^{r(A)}k\beta)$$

where

$${}^i\alpha = ({}^i\alpha_1(x_1, x_2, \dots, x_s), {}^i\alpha_2(x_1, x_2, \dots, x_s), \dots, {}^i\alpha_r(x_1, x_2, \dots, x_s)) \in K[x_1, x_2, \dots, x_s]^r,$$

$${}^i\beta = ({}^i\beta_1(x_1, x_2, \dots, x_s), {}^i\beta_2(x_1, x_2, \dots, x_s), \dots, {}^i\beta_s(x_1, x_2, \dots, x_s)) \in K[x_1, x_2, \dots, x_s]^s.$$

Alice substitutes p_i instead of variables x_i and gets coordinates ${}^i\alpha(p_1, p_2, \dots, p_s) = d(i)$ and ${}^i\beta(p_1, p_2, \dots, p_s) = e(i)$. So she forms sequence of colours

$$(d(1), e(1), d(2), e(2), \dots, d(r(A)k), e(r(A)k))$$

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for the walk in the graph $I(K)$ starting in the point $p = (p_1, p_2, \dots, p_s, p_{s+1}, \dots, p_{s+m})$.

Alice computes the vertices of this walk as

$$v_1 = N_d(1)(p), u_1 = N_e(1)(v_1),$$

$$v_2 = N_d(2)(u_1), u_2 = N_e(2)(v_2),$$

$$v_3 = N_d(3)(u_2), u_3 = N_e(3)(v_3),$$

$\dots,$

$$v_{r(A)k} = N_d(r(A)k)(u_{r(A)(k-1)}), u_{r(A)k} = N_e(r(A)k)(v_{r(A)k}) = u_A.$$

She applies $G(B)$ consecutively $k(A)$ times to construct

$$G(B)(u_A) = {}^1u_A, G(B)({}^1u_A) = {}^2u, \dots, G(B)({}^{k(A)-1}u_A) = {}^{k(A)}u_A = v_A.$$

Finally Alice takes the symbolic word

$$({}^1w^*)^{r(A)} = ({}^1\mu, {}^1\gamma, {}^2\mu, {}^2\gamma, \dots, {}^{r(A)k}\mu, {}^{r(A)k}\gamma).$$

She specialize variables $x_i, i = 1, 2, \dots, s$ as first s coordinates of the point v_A and gets the sequence of colours $(d^*(1), e^*(1), d^*(2), e^*(2), \dots, d^*(r(A)k), e^*(r(A)k))$ of the walk in the graph $I(K)$ with the starting point v_A .

Alice computes the vertices of this walk as

$$v_1^* = N_d^*(1)(v_A), u_1^* = N_e^*(1)(v_1^*),$$

$$v_2^* = N_d^*(2)(u_1^*), u_2^* = N_e^*(2)(v_2^*),$$

$$v_3^* = N_d^*(3)(u_2^*), u_3^* = N_e^*(3)(v_3^*),$$

$\dots,$

$$v_{r(A)k}^* = N_d^*(r(A)k)(u_{r(A)(k-1)}^*), u_{r(A)k}^* = N_e^*(r(A)k)(v_{r(A)k}^*) = v(A).$$

She can use the vector $c = (v(A)_s + 1, v(A)_s + 2, \dots, v(A)_s + m)$ as the collision element of the protocol Bob computes the collision tuple c in his own way. He takes the word

$${}^1w^{r(B)} = ({}^1A, {}^1B, {}^2A, {}^2B, \dots, {}^kA, {}^kB)^{r(B)} = ({}^1\alpha', {}^1\beta', {}^2\alpha', {}^2\beta', \dots, {}^{r(B)k}\alpha', {}^{r(B)k}\beta'),$$

substitutes p_i instead of variables $x_i, i = 1, 2, \dots, s$ into coordinates of the symbolic walk and gets the sequence of colours $(d'(1), e'(1), d'(2), e'(2), \dots, d'(r(B)k), e'(r(B)k))$ for the walk in the graph $I(K)$ starting in the point $p = (p_1, p_2, \dots, p_s, p_{s+1}, \dots, p_{s+m})$.

Bob computes the vertices of this walk as

$$v'_1 = N_d'(1)(p), u'_1 = N_e'(1)(v'_1),$$

$$v'_2 = N_d'(2)(u'_1), u'_2 = N_e'(2)(v'_2),$$

$$v'_3 = N_d'(3)(u'_2), u'_3 = N_e'(3)(v'_3),$$

$\dots,$

$$v'_{r(B)k} = N_d'(r(B)k)(u'_{r(B)(k-1)}), u'_{r(B)k} = N_e'(r(B)k)(v'_{r(B)k}) = u_B.$$

He applies $G(A)$ consecutively $k(B)$ times to construct

$$G(A)(u_B) = {}^1u_B, G(A)({}^1u_B) = {}^2u, \dots, G(A)({}^{k(B)-1}u_B) = {}^{k(B)}u_B = v_B.$$

Finally Bob takes the symbolic word

$$({}^1w')^{r(B)} = ({}^1\mu', {}^1\gamma', {}^2\mu', {}^2\gamma', \dots, {}^{r(B)k}\mu', {}^{r(B)k}\gamma').$$

He specialize variables $x_i, i = 1, 2, \dots, s$ as first s coordinates of the point v_B and gets the sequence of colours $(d''(1), e''(1), d''(2), e''(2), \dots, d''(r(B)k), e''(r(B)k))$ of the walk in the graph $I(K)$ with the starting point v_B . Bob computes the vertices of this walk as

$$\begin{aligned} v_1'' &= N_d''(1)(v_B), u_1'' = N_e''(1)(v_1''), \\ v_2'' &= N_d''(2)(u_1), u_2'' = N_e''(2)(v_2''), \\ v_3'' &= N_d''(3)(u_2^*), u_3'' = N_e''(3)(v_3''), \\ &\dots, \\ v_{r(B)k}'' &= N_d''(r(B)k)(u_{r(B)(k-1)}''), u_{r(B)k}'' = N_e''(r(B)k)(v_{r(B)k}'') = v(B). \end{aligned}$$

He can use the vector $c = (v(B)_{s+1}, v(B)_{s+2}, \dots, v(B)_{s+m})$ as the collision element of the protocol. The output of proposed above protocols based on linguistic graph is the tuple of ring elements. This form is convenient for the key establishment.

4 On the implementation with homomorphic images of infinite forest.

Below we consider the case of linguistic graphs $I_n(K)$ of type $(1, 1, n-1)$. The semigroup ${}^{1,1}S(K)$ consists of symbolic words of kind $(A_1(x_1), B_1(x_1), A_2(x_1), B_2(x_1), \dots, A_K(x_1), B_K(x_1))$ where A_i and B_i are polynomials in single variable x_1 . We consider the subsemigroup ${}^{1,1}S_n(K)$ which consists of symbolic words as above with A_i and B_i of kind $x_i + a_i$ and $x_i + b_i$ where a_i and b_i are constants. We will encode the symbolic word as tuple $(a_1, b_1, a_2, b_2, \dots, a_k, b_k)$ of even length from the totality $F(K)$ of sequences of elements of K of finite length. We introduce the product of tuples from $F(K)$ accordingly the following rule

$$(a_1, a_2, \dots, a_r) \times (b_1, b_2, \dots, b_s) := (a_1, a_2, \dots, a_r, b_1 + a_r, b_2 + a_r, \dots, b_s + a_r)$$

and present the the reverse $^*(a_1, a_2, \dots, a_r)$ of $(a_1, a_i, \dots, a_r)$ as

$$^*(a_1, a_2, \dots, a_r) := (a_{r-1} - a_r, a_{r-2} - a_r, \dots, a_1 - a_r, -a_r).$$

The restrictions of these operations on $F'(K) = {}^{1,1}S(K)$ coincide with the already defined operations of composition and taking reverse in this semigroup.

The values of already introduced homomorphism ${}^n\Psi(I_n(K))$ of the semigroup ${}^{1,1}S_n(K)$ into Cremona affine semigroup $\text{End } K[x_1, x_2, \dots, x_n]$ defined by the linguistic graph $I_n(K)$ into Cremona affine semigroup $\text{End } K[x_1, x_2, \dots, x_n]$ defined by the linguistic graph $I_n(K)$ will be computed via the following procedure. We take the $(c_1, c_2, \dots, c_r) \in {}^{1,1}S_n$ and form path

$$(x_1, x_2, \dots, x_n)I[x_1 + c_1, {}^1f_2, {}^1f_3, \dots, {}^1f_n]I(x_1 + c_2, {}^2f_2, \dots, {}^2f_n)I \dots (x_1 + c_r, {}^rf_2, \dots, {}^rf_n).$$

The multivariate map H sending $(x_1, x_2, \dots, x_n)$ to $(x_1 + c_2r, {}^2rf_2, \dots, {}^2rf_n)$ is the value of the homomorphism on selected symbolic walk. The Protocol 1 can be implemented in the case of linguistic graph of type $1, 1, n-1$. We select for the implementation known linguistic graph $D(n, K)$ defined over arbitrary commutative ring K .

These graphs are bipartite with set of vertices $V = PUL, |P \cap L = 0|$. A subset of the vertices P is called the set of points and another subset L is called the set of lines. Let P and L be two copies of Cartesian power K^n , where $n \geq 2$ is an integer. Two types of brackets are used in order to distinguish points from lines. It has a set of vertices (collection of points and lines), which are n -dimensional vectors over

$$K : (p) = (p_1, p_2, p_3, p_4, \dots, p_i, p_{i+1}, p_{i+2}, p_{i+3}, \dots, p_n),$$

$$[l] = [l_1, l_2, l_3, l_4, \dots, l_i, l_{i+1}, l_{i+2}, l_{i+3}, \dots, l_n].$$

307 The point (p) is incident with the line $[l]$, if the following relations between their coordinates hold:

$$308 \quad l_2 - p_2 = l_1 p_1,$$

$$309 \quad l_3 - p_3 = l_2 p_1,$$

$$310 \quad l_4 - p_4 = l_1 p_2,$$

$$311 \quad l_i - p_i = l_1 p_{i-2},$$

$$312 \quad l_{i+1} - p_{i+1} = l_{i-1} p_1$$

$$313 \quad l_{i+2} - p_{i+2} = l_i p_1,$$

$$314 \quad l_{i+3} - p_{i+3} = l_1 p_{i+1}$$

316 where $i \geq 5$. Connected component of edge-transitive graph $D(n, F_q) = D(n, q)$ is denoted by
 317 $CD(n, q)$ [22]. Notice that all connected components of the natural projective limit $D(q)$ of graphs
 318 $D(n, q), n \rightarrow \infty$ are q -regular trees.

319 The following result on the symbolic transformations corresponding to $D(n, K)$ are known.

320 ► **Lemma 6.** [38]. *Let K be commutative ring. Then ${}^n\Psi(D(n, K))$ is stable group of degree 3.*

321 ► **Lemma 7.** [38] *Let K be commutative ring. Then the density of ${}^n\Psi(D(n, K))$ is $O(n)$. This fact*
 322 *is supported by the results of computer simulations.*

323 Let us consider the implementation of the Protocol 1 in the case of graphs $D(n, K)$ and evaluate its
 324 complexity.

325 4.1 Protocol 2

326 Alice and Bob choose the finite commutative ring K . They share the pseudorandom or genuinely
 327 random sequence $c_1, c_2, \dots, c_r, b_1, b_2, \dots, b_{t-1}$ of elements from K^j where r is even number and $t - 1$
 328 is odd integer. They set $b_t = 0$.

329 We assume that $c_i \neq c_{i+2}, i = 1, 2, \dots, r - 2$ and $b_i \neq b_{i+2}, i = 1, 2, \dots, t$.

330 Correspondents select the parameter n and work with publicly known graph $D(n, K)$. They
 331 agree on pseudorandom sequence $d = (d_1, d_2, \dots, d_n)$. This parameter is known publicly.

332 They treat $c = (c_1, c_2, \dots, c_r)$ and $b = (b_1, b_2, \dots, b_t)$ as elements of semigroup $(F'(K), \times)$. Alice
 333 selects her private parameters $r(A)$ and $k(A)$. She computes $c^{r(A)} b^{k(A)} (*c)^{r(A)} = x(A)$ and computes
 334 ${}^n\Psi(x(A)) = g(A)$ in its standard form.

335 She sends $g(A)$ to Bob via the open channel. She computes ${}^n\Psi(c) = h$ and ${}^n\Psi(*c) = h^{-1}$ for
 336 herself. Note that $g(A)$ sends x_1 to itself. Bob selects his private parameters $r(B)$ and $k(B)$. He
 337 computes $c^{r(B)} b^{k(B)} (*c)^{r(B)} = x(B)$ and computes ${}^n\Psi(x(B)) = g(B)$ in its standard form. He sends
 338 $g(B)$ to Alice via the open channel and computes h and h^{-1} as well.

339 Alice takes polynomial map h and computes ${}^1d = h(d), {}^2d = h({}^1d) \dots, {}^{r(A)}d = h({}^{r(A)-1}d) = d'$.
 340 She computes ${}^1b = g(B)d', {}^2b = g(B)({}^1b), \dots, {}^{k(A)}b = g(B)({}^{k(A)-1}b) = d''$. Alice takes h^{-1} and
 341 forms ${}^1c = h^{-1}(d''), {}^2c = h^{-1}({}^1c), \dots, {}^{r(A)}c = h^{-1}({}^{r(A)-1}c) = Y(A)$.

Bob takes map h and computes ${}^1d = h(d), {}^2d = h({}^1d) \dots, {}^{r(B)}d = h({}^{r(B)-1}d) = v$. He computes

$${}^1u = g(A)v, {}^2u = g(A)({}^1u), \dots, {}^{k(B)}v = g(A)({}^{k(A)-1}v) = w.$$

Bob takes h^{-1} and forms

$${}^1z = h^{-1}(w), {}^2z = h^{-1}({}^1z), \dots, {}^{r(B)}c = h^{-1}({}^{r(B)-1}z) = Y(B).$$

342 So correspondents have the collision vector $Y(A) = Y(B) = (d_1, y_2, \dots, y_n)$. They can use vector
 343 $(y_2, y_3, \dots, y_n)$ as the established key. We assume that parameters $r(A), r(B), k(A), k(B), r$ and t
 344 are of size $O(1)$. The computation of the value of ${}^n\Psi$ on the symbolic word of length $O(1)$ takes
 345 time $O(n)$. The protocol requires $O(1)$ calculations of the value of ${}^n\Psi$ on the words of finite length.
 346 of $(y_2, y_3, \dots, y_n)$ requires $O(1)$ calculations the value of $h, h^{-1}, g(A)$ or $g(B)$ on the element from
 347 K^n . Each of these transformations has density $O(n)$. So the value can be computed in time $O(n^2)$.
 348 It means that the execution time of the protocol is $O(n^2)$.

► **Remark 8.** Adversary has to compute $g = n^\Psi, h$ and h^{-1} . He/she has to solve for y the equation $h^y y g^x = h^{-y} = g(A)$ (or $g(B)$) in the group of cubic transformations induced by walks on the space of point. This problem in the case of elements of affine Cremona group is postquantum intractable. Alternatively adversary is able to use his knowledge on the graph $D(n, K)$ and try to find the shortest pass between $(d_1, d_2, \dots, d_n)$ and $(y_1, y_2, \dots, y_n)$ with use of Dijkstra algorithm or its modification. This option is unfeasible because the order v of the graph is $2q^n$, where q is the cardinality of the finite ring K and the complexity of Dijkstra procedure is $v \log(v)$.

► **Remark 9.** In fact correspondents can compute the values of h and h^{-1} in time $O(n)$ because of their knowledge on tuples c and jc . They can apply consecutively operators $N_{c_1}, N_{c_2}, \dots, N_{c_r}$ to point $p = (p_1, p_2, \dots, p_n)$ and get $h(p)$. This possibility does not affect general complexity $O(n^2)$ because of the necessity to compute values of $g(A)$ or $g(B)$.

4.2 Protocol 3. Hidden graph protocol

Assume that Alice has the data of previous protocol, i.e. parameter n , finite commutative ring K , graph $D(n, K)$ and elements $c = (c_1, c_2, \dots, c_r)$ and $(b_1, b_2, \dots, b_t)$ for even $r, r > 0$ and even parameter t . The requirement that $b_t = 0$ is immaterial. Additionally she selects element L of $AGL_n(K)$ of density $O(1)$ such that its inverse also has the density $O(1)$. She calculates the standard forms of elements $H = L^n \Psi(c) L^{-1}, H^{-1}$ and $G = L^n \Psi_I(b) L^{-1}, I = D(n, K)$ from the subgroup $X = \langle HG \rangle$ of $L^n \Psi(^{1,1}S_n(K)) L^{-1} = Y$. Note that each element of the group X has degree 3 and density $O(n)$ accordingly Lemma 6 and 7. Alice sends the standard forms of G and H to Bob via the open channel.

Alice selects parameters $r(A)$ and $k(B)$. She can use her knowledge on elements c and b from $(n$ -the semigroup $^{1,1}S_n(K)$ and computes $c^{r(A)} b^{k(A)} (*c)^{r(A)} = e$ and $\Psi(e)(e) = g(A)$ in time $O(n)$. She sends $g(A)$ to Bob via the open channel. Bob chooses his own parameter $r(B)$ and $k(B)$ and computes $g(B)$ as $Y = H^{r(B)} G^{k(B)} H^{-r(B)}$. Note that he need $O(1)$ operations in X . The multiplication of cubic elements of degree 3 of density $O(n)$ requires $O(n^5)$ elementary operations. He sends $G(B)$ to Alice. Alice can compute the collision element as $Y = H^{r(A)} G^{k(B)} H^{-r(A)}$ in time $O(n^5)$ as well.

► **Remark 10.** Alice and Bob can use action of X on K^n and conduct the protocol for transformation group but it does not change the complexity $O(n^5)$ because the necessity to compute $g(B)$.

► **Remark 11.** In the case of finite fields of characteristic 2 Alice can use general element L from $AGL_n(K)$ of density $O(n)$. In this case the density of element from subgroup X will be $O(n^2)$. This estimation is obtained via computer simulations. It means that the complexity of group operations will be $O(n^9)$.

Let us compare the Protocol 3 with the algorithms based on the Navigation Problem.

In paper [38] platform Y were used for the implementation of several algorithms based on the navigation problem, i.e. problem of finding the decomposition

$$(g_i(1))^{k(1)} (g_i(2))^{k(2)} \dots (g_i(s))^{k(s)}$$

of element $g \in Y$ given in its standard form and taken from the subgroup $G = \langle g_1, g_2, \dots, g_s \rangle$ of Y generated by known elements g_i .

We select hidden tame homomorphism protocol on the platform Y for the comparison with the Protocol 2 of previous section. The general scheme of the protocol is the case of the subsemigroup of affine Cremona group is the following one.

Let S be the subsemigroup of ${}^nCS(K)$ and Φ be homomorphism of S into ${}^mCS(K)$ where $n > m$. Assume that $S' = Im(\Phi)$. Alice selects elements $a_1, a_2, \dots, a_k$ of S and computes their homomorphic images $b_i = \Phi(a_i), i = 1, 2, \dots, k$. She takes elements $h_1 \in {}^nCG(K)$ and $h_2 = {}^mCG(K)$ outside of the normalisers S and S' and calculates the standard forms of $c_i = h_1 a_i h_1^{-1}$ and $d_i = h_2 b_i h_2^{-1}$. Alice sends pairs of kind (c_i, d_i) to Bob

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Bob selects sequence $I(1), i(2), \dots, i(r), r \geq 2$ such that $i(j) \neq i(j+1)$ for $j = 1, 2, \dots, r-1$ and positive integers $s(1), s(2), \dots, s(k)$. He calculates the standard form of $c_i(1)^k(1)c_i(2)^k(2) \dots c_i(r)^k(r) = g(B)$ and sends it to Alice. Bob computes the collision element g as the standard form of $d_i(1)^k(1)d_i(2)^k(2) \dots d_i(r)^k(r)$.

Alice computes $h_1^{-1}g(B)h_1 = g(A)$. She computes $\Phi(g(A) = g'(A)$ and collision element g as $(h_2)^{-1}g'(A)h_2$.

In paper [38] this protocol with implemented in case of platform Y and its special homomorphism induced by the automorphism of graph $D(n, K)$ onto graph $A(n, K)$. Elements h_1 and h_2 were taken from $AGL_n(K)$ In the case of elements L, h_1 and h_2 and their inverses of density $O(1)$ the complexity of this algorithm is $O(n^5)$. If these elements have density $O(n)$ then the complexity of this algorithm is $O(n^9)$.

5 The results of computer simulations

Assume that parameters $r(A), K(A), r(B)$ and $K(B)$ are of size $O(1)$ and length of tuples b and c are $O(n)$,

Noteworthy that Alice and Bob form the tuples $w(A) = c^{r(A)}b^{k(A)}(*c)^{r(A)}$ and $W(B) = c^{r(B)}b^{k(B)}(*c)^{r(A)}$ in time $O(n)$. The complexity of algorithm for the correspondents depends on the parameters $l(A) = 2l(c)r(A) + k(A)l(b)$ and $l(B) = 2l(c)r(B) + k(B)l(b)$ where $l(c)$ and $l(b)$ are length of c and b .

We assume that m is one of these parameters and the protocol is set for graph $D(n, q)$. So correspondent create cubic transformation F of $(F_q)^n$. We present the global density $g(m, q)$ of T . Note that the density $den(F)$ of F can be evaluated as $g(m, q)/n$.

Case $F_{2^{24}}$

Table 1 Total number of monomial terms of the cubic map induced by the graph

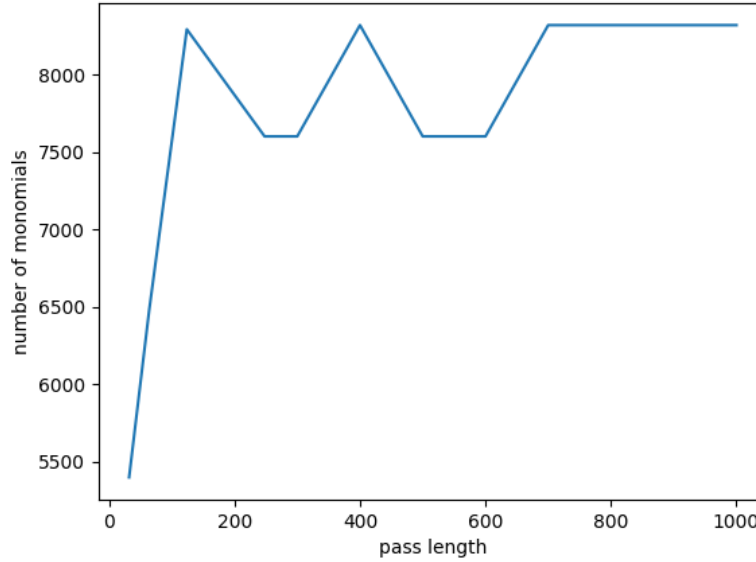
	Pass length $l(A)$			
# of variables x_i	32	64	124	248
16	112	47	145	112
32	456	288	545	462
64	1815	1456	2113	1882
128	5399	6479	8321	7602
256	12567	20047	32988	30286

Table 2 Generation time for the map (ms),

	Pass length $l(A)$			
# of variables x_i	32	64	124	248
16	21,9	37,5	68,6	134
32	68,8	131	282,6	606,6
64	244,2	522,4	1653,3	2673,3
128	544,6	2577	7191,4	140202,6
256	1193,6	5555,7	24337,1	62329,6

We have written a program for generating elements and for encrypting a text using the generated public key. The program is written in SAGE and Python. We used an MacBook with a Intel Core 1,2 GHz processor, 8GB RAM, and the WmaxOS Sierra operating system. The generation time for the map is presented in the Table 2.. The Table 1 and Figure 1 demonstrates the stabilization of global density on some asymptotic value. Thus the adversary is not able to get the information

■ **Figure 1** Number of monomial terms of the cubic map induced by the graph, $n = 128, F_{2^{24}}$



on the parameter m (and corresponding integers $r(A)$, $k(A)$ or $r(B)$, $k(B)$ from his knowledge on global density. Noteworthy that values of $r(A)$, $k(A)$, $r(B)$, $k(B)$ can be negative as well. Negative value means that correspondent uses reverse word instead of c or b .

6 Conclusion

Development of efficient key establishment protocols with the postquantum resistance to adversarial attacks is active area of research (see [43]-[48]). Demand of efficient standards for these protocols is especially important for the area of electronic public verifiable voting [50]-[52].

We present efficient graph based key exchange protocol defined in terms of well known family of bipartite graphs of $D(n, q)$. The partition sets of the graph are vector spaces $(F_q)^n$ and the incidence is defined via the system of quadratic equations. The walks of even length on this graph generate large group of cubic polynomial transformations acting on each partition sets. It can be used as the standard platform for the implementation of the scheme of Twisted Diffie-Hellman protocol for transformation groups. The adversary has to solve the Conjugacy Power Problem (CPP) for elements of the group given in their standard forms of multivariate transformation or use Dijkstra algorithm to study the walks between the tuple on $(F_q)^n$ and its value for the polynomial map sent by one of the correspondents via open channel. The last option is unfeasible because the complexity of Dijkstra algorithm is $v \log(v)$ where v is order of the investigated graph which is $2q^n$ in the case of the used graph. The polynomial algorithm to solve the CPP in case of cubic automorphisms of $F_q[x_1, x_2, \dots, x_n]$ is unknown.

These arguments justify the security of the protocol with the execution complexity $O(n^2)$ in the case of symbolic words of length $O(n)$ and private integers parameters of size $O(1)$. In the case of words of length $O(1)$ and private parameters of size $O(n)$ the complexity remains the same. If words have length $O(n)$ and integers have size $O(n)$ the complexity is $O(n^3)$. It compares well with the complexity of protocols of Noncommutative Cryptography implemented with platform of Eulerian transformations of $K[x_1, x_2, \dots, x_n]$ (see [49] and further references). Recall that Eulerian transformation is sending variable x_i into the single monomial term $(x_1)^a(1, i)(x_2)^a(2, i) \dots (x_n)^a(n, i)$. The complexity of semigroup operation is $O(n^3)$. So the complexity of the protocols is $O(n^3)$.

We implement the protocol in the case of fields $F_q, q = 2^m, m = 8, 16, 24$. Computer simulations demonstrate that densities of cubic maps created by Alice and Bob do not depend on the choice of the private keys. There is a possibility to hide the graph via the conjugation of the group platform with element L from $AGL_n(F_q)$. This obfuscation does not allow adversary to conduct graph based attacks but it increases the execution complexity which is $O(n^5)$ even in case when densities of L and L^{-1} are $O(1)$. In case of the L in general position the complexity will be $O(n^9)$, which coincides with the complexity of protocols based on Navigation Problem on the same platform (see [38] and further references). Mentioned above protocols are defined in more general case of graphs $D(n, K)$ where K is arbitrary finite commutative ring. There is a possibility to change graph $D(n, K)$ for other Jordan-Gauss graphs over K defined in [42].

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